Physics 23: Electromagnetism and Waves

Practice Midterm Examination

Spring 2026 — Purcell Chapters 1 & 2

Total: **100 points** Time allowed: 75 minutes

Instructions:

- Show all work clearly. Box your final answers.
- State all symmetry arguments *before* applying Gauss's law or integrating.
- You may use $k = 1/(4\pi\varepsilon_0) \approx 9 \times 10^9\ \mathrm{N\,m^2/C^2}$.
- Scratch work without clear reasoning earns no credit.

Problem 1

[18 points]

Reading Vector-Field Diagrams

The six panels below (Purcell Figure 2.39) show two-dimensional vector fields in the xy-plane. For each panel, answer the three questions in the table:

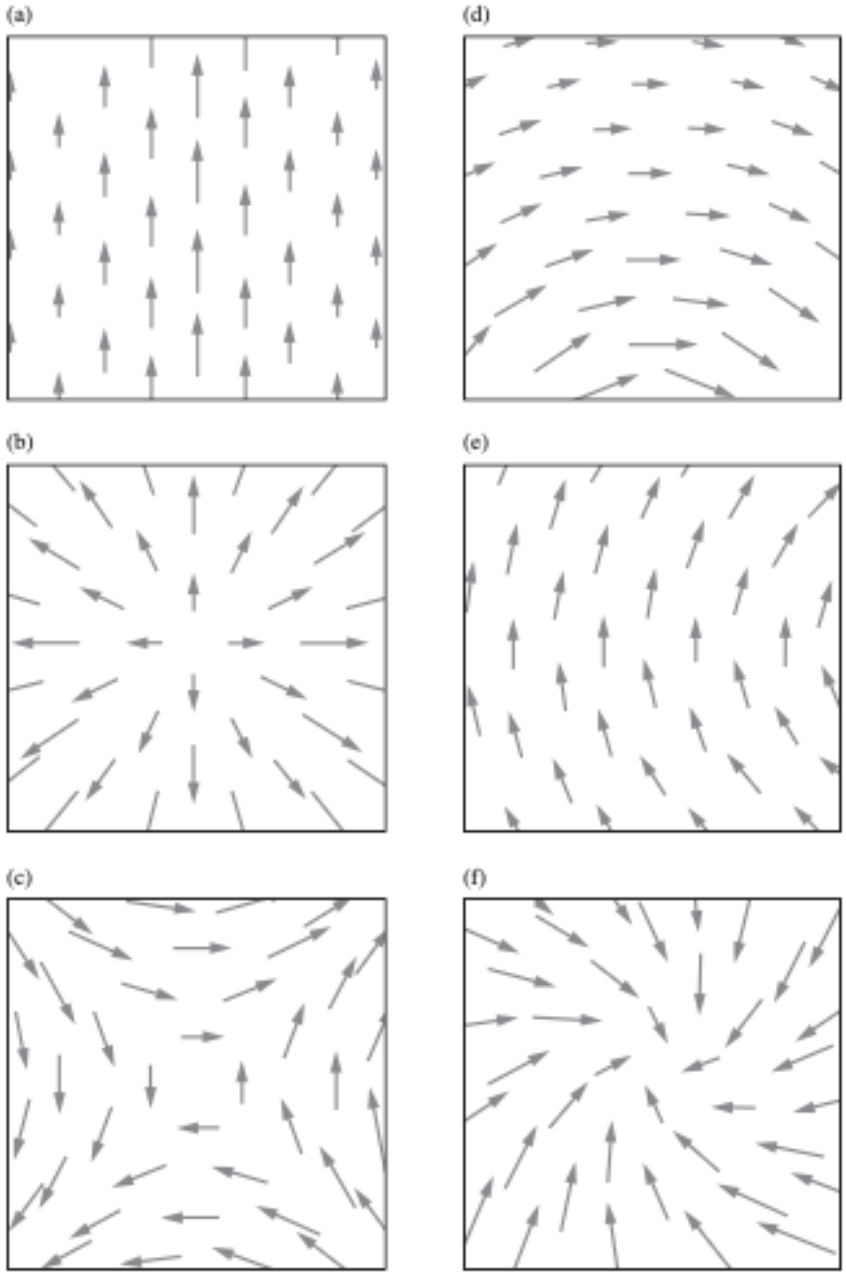
(i) Is $\nabla \cdot \vec{E} = 0$ in the region shown?

(ii) Is $\nabla \times \vec{E} = \vec{0}$ in the region shown?

(iii) Could this field be a *static electric field in a charge-free region*?

Give a one-sentence physical or mathematical justification for each answer.

Hint: Divergence $\Leftrightarrow$ net flux out of a tiny closed surface (field lines start/end). Curl $\Leftrightarrow$ net circulation around a tiny closed loop.


(a)
(d)
(b)
(e)
(c)
(f)

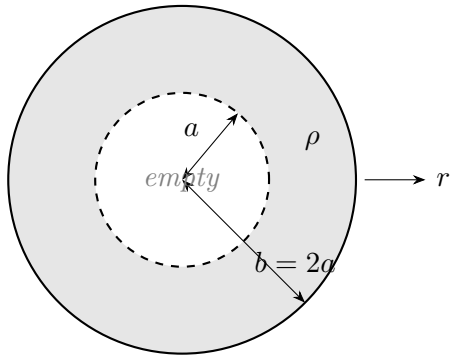
Panel	**(i)** $\nabla \cdot \vec{E} = 0$**?**	**(ii)** $\nabla \times \vec{E} = \vec{0}$**?**	**(iii) Electrostatic in charge-free region?**
(a)			
(b)			
(c)			
(d)			
(e)			
(f)			

Problem 2

[30 points]

Thick Spherical Shell: Gauss's Law to Potential

An insulating thick spherical shell has inner radius a and outer radius $b = 2a$. The shell carries uniform volume charge density $\rho > 0$; the cavity $r < a$ is empty.



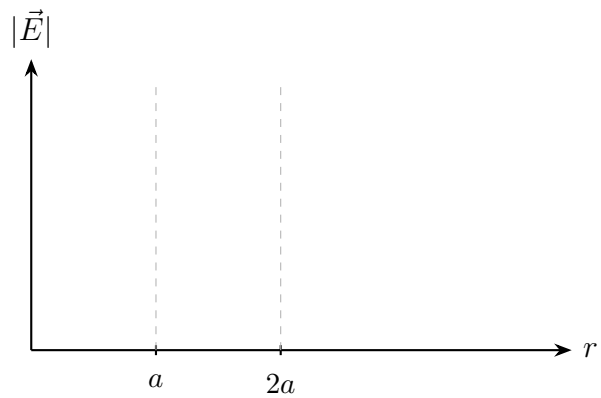
(a) **[4 pts] Symmetry argument.** Without calculation, explain why $\vec{E}(r)$ must point in the $\hat{r}$ direction and depend only on r. State the specific symmetries of the charge distribution you are using.

(b) **[10 pts] Electric field via Gauss's law.** Draw an appropriate Gaussian surface for each region, compute the enclosed charge, and find $\vec{E}(r)$ in all three regions. Express answers in terms of ρ, a, r, and ε_0. Then **sketch** $|\vec{E}|$ versus r on the axes below, labeling the values at $r = a$ and $r = 2a$.

(i) $r < a$: $\vec{E} =$ _____

(ii) $a \leq r \leq 2a$: $\vec{E} =$ _____

(iii) $r > 2a$: $\vec{E} =$ _____



(c) **[4 pts] Electric flux.** Compute the total flux Φ_E through a spherical Gaussian surface at $r = \frac{3}{2}a$. First use your field from (b); then verify by computing Q_{enc} directly and applying Gauss's law. Both methods must agree.

(d) **[8 pts] Electric potential.** Taking $\varphi(\infty) = 0$, find $\varphi(r)$ in all three regions by integrating inward from infinity:

$$\varphi(r) = -\int_{\infty}^{r} \vec{E} \cdot d\vec{\ell}.$$

Work region by region, starting from $r > 2a$, and *check continuity* of φ at each boundary.

(i) $r > 2a$:

(ii) $a \leq r \leq 2a$:

(iii) $r < a$:

(e) **[4 pts] Limits and interpretation.**

(i) Show that for $r \gg 2a$ your potential reduces to $\varphi(r) \approx \dfrac{Q_{\text{tot}}}{4\pi\varepsilon_0 r}$, where Q_{tot} is the total charge of the shell. Express Q_{tot} in terms of ρ and a. What does this limit mean physically?

(ii) Compute $\varphi(0)$. Is it greater than, equal to, or less than $\varphi(a)$? Why? (No new integral needed—use what you found in part (d).)

Problem 3

[20 points]

Energy Stored in a Uniformly Charged Sphere

A solid sphere of radius R carries uniform volume charge density ρ. The electric field (which you have derived in lecture and homework) is:

$$\vec{E}(r) = \begin{cases} \dfrac{\rho\, r}{3\varepsilon_0}\,\hat{r} & r < R, \\ \dfrac{\rho R^3}{3\varepsilon_0 r^2}\,\hat{r} & r > R. \end{cases}$$

(a) **[4 pts]** Write the electrostatic energy density $u_E(r) = \dfrac{\varepsilon_0}{2}E^2$ explicitly for $r < R$ and for $r > R$.

(b) **[12 pts]** Compute the total energy stored in the electric field:

$$U = \int_{\text{all space}} u_E\, dV = \int_0^\infty u_E\,(4\pi r^2)\, dr\,.$$

Split the integral at $r = R$. Carry out both integrals and show that

$$\boxed{U = \frac{3Q^2}{20\pi\varepsilon_0 R}},$$

where $Q = \frac{4}{3}\pi R^3\rho$ is the total charge. Show all steps clearly.

(c) **[4 pts] Sanity checks and physical insight.**

(i) Verify the units of $U = 3Q^2/(20\pi\varepsilon_0 R)$. Show explicitly that $Q^2/(\varepsilon_0 R)$ has units of joules.

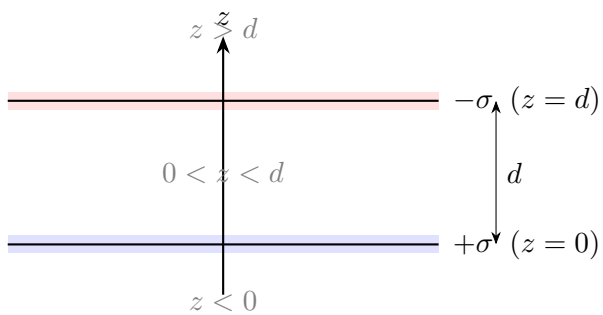
(ii) As $R \to 0$ with Q fixed (a classical point charge), what happens to U? What does this imply about the self-energy of a point charge in classical physics?

Problem 4

[17 points]

Two Infinite Parallel Sheets

Two infinite, flat sheets of charge lie in the xy-plane. The bottom sheet (at $z = 0$) has surface charge density $+\sigma$; the top sheet (at $z = d$) has surface charge density $-\sigma$.



(a) **[5 pts] Electric field by superposition.** Recall that a single infinite sheet with surface charge density σ produces $E = \sigma/(2\varepsilon_0)$ pointing away from the sheet on each side. Using superposition, find $\vec{E}$ in all three regions: $z < 0$, $0 < z < d$, and $z > d$.

(b) **[4 pts] Electric potential.** Set $\varphi(0) = 0$ (the bottom sheet is the reference). Integrate $\vec{E}$ to find $\varphi(z)$ in all three regions. Check that φ is continuous at $z = 0$ and at $z = d$.

(c) [**4 pts**] **Field energy between the plates.** Compute the energy stored per unit area between the plates:

$$\frac{U}{A} = \int_0^d u_E \, dz = \int_0^d \frac{\varepsilon_0}{2} E^2 \, dz \, .$$

(d) [**4 pts**] **Physical interpretation.**

(i) What is $\vec{E}$ *outside* both sheets (i.e., for $z < 0$ and $z > d$)? Explain why this is expected from the perspective of superposition.

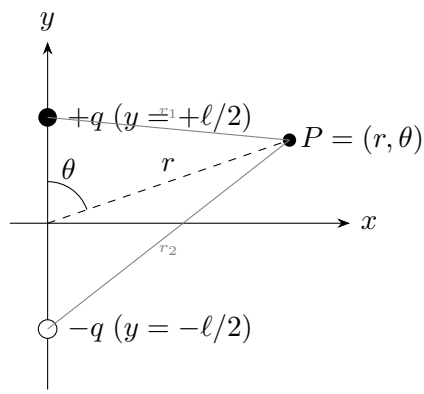
(ii) In the limit $d \to 0$ with σ fixed, what happens to the field energy per unit area U/A? Is this physically reasonable? Briefly explain why the field energy vanishes in this limit.

Problem 5

[15 points]

The Electric Dipole

Two point charges $+q$ (at $y = +\ell/2$) and $-q$ (at $y = -\ell/2$) form a dipole with dipole moment $p = q\ell$. A field point P is at distance r from the midpoint, at polar angle θ measured from the $+y$ axis (the axis joining the charges).



(a) **[3 pts]** Write the *exact* potential φ_P at point P due to both charges, in terms of q, ε_0, r_1 (distance from $+q$ to P), and r_2 (distance from $-q$ to P).

(b) **[5 pts]** In the **far-field limit** $r \gg \ell$, use the approximations

$$r_1 \approx r - \tfrac{\ell}{2}\cos\theta, \qquad r_2 \approx r + \tfrac{\ell}{2}\cos\theta, \qquad \frac{1}{r_{1,2}} \approx \frac{1}{r}\left(1 \pm \frac{\ell\cos\theta}{2r}\right)$$

to show that

$$\varphi(r,\theta) \;\approx\; \frac{p\cos\theta}{4\pi\varepsilon_0\, r^2}\,, \qquad p = q\ell\,.$$

Show all algebra clearly.

(c) **[4 pts]** Using $\vec{E} = -\nabla\varphi$ in polar coordinates,

$$\vec{E} = -\frac{\partial\varphi}{\partial r}\,\hat{r} - \frac{1}{r}\frac{\partial\varphi}{\partial\theta}\,\hat{\theta},$$

derive the far-field electric field $\vec{E}(r,\theta)$. Your final answer should take the form

$$\vec{E}(r,\theta) = \frac{p}{4\pi\varepsilon_0 r^3}\big(A\cos\theta\,\hat{r} + B\sin\theta\,\hat{\theta}\big)$$

for some integer coefficients A and B. Find A and B.

(d) **[3 pts]** Evaluate $\vec{E}$ at (i) $\theta = 0$ (on the $+y$ axis, far above $+q$) and (ii) $\theta = \pi/2$ (on the equatorial x-axis). In each case state the direction of $\vec{E}$ and verify it makes physical sense by sketching the expected direction of force on a positive test charge placed at that location.

End of Practice Midterm. Good luck on the real exam!